2026 Greenbrier Science Olympiad Invitational

January 10, 2026

Disease Detectives C

EXAM

Please record your responses on the provided answer sheet! Answers on the exam book will not be scored!
Point values for each question or set of questions are listed in [brackets].

There are 111 points available to earn from 86 total questions.
TIME TO COMPLETE: 50 minutes
Please write out answers to calculation-based questions using two decimal places.

GOOD LUCK!

TOURNAMENT STAFF SCORING USE ONLY

Part 1: _____ Part 2: _____ Part 3: _____ Part 4: _____ **TOTAL**: _____

Part 1 | Definitions and background knowledge

Briefly define the following terms. Complete sentences are not required. [**1 each**]

1. Endemic
2. Epizootic
3. Zoonotic disease
4. Nosocomial
5. Virulence
6. Fomite
7. Pandemic
8. Specificity
9. Vertical transmission
10. Basic reproductive number (R_0)

What do these acronyms stand for? [**1 each**]

11. HIV
12. CDC
13. MSF
14. SARS
15. STEC
16. HIPAA
17. COVID
18. MMWR
19. Tdap
20. MMR

21. Arrange these disease outbreaks in the correct chronological order. [**3**]
    a. Swine flu (Influenza A virus subtype H1N1) pandemic
    b. Black death
    c. Broad Street cholera outbreak in Soho, London
    d. Justinian plague
    e. COVID-19 pandemic
    f. Spanish flu (Influenza A virus subtype H1N1) pandemic
    g. Legionnaire's disease outbreak from drinking water in Flint, Michigan

Using the word bank provided below, fill in each blank with one word or term from the word bank. Some terms may be used once, more than once, or not at all. [**1 per correct blank**]

Active	Endemic	Index case	Smallpox
Carrier	GRADE	Inoculation	Syndromic
CDC	Holoendemic	Latent period	Vaccination
Chain of infection	Hyperendemic	Observational	Vector
Effectiveness	Immunization	Pathogenicity	Virulence
Efficacy	Incubation period	Red cross	WHO

22. The _____ cannot be completed if even one of its six components are missing.
23. Washing your hands regularly can protect against transmission of disease from a _____, like doorknobs and elevator buttons, which are often encountered in everyday life.
24. _____ refers to the ability of an organism to cause disease, while _____ refers to the severity of disease caused by an agent.
25. Vaccination can disrupt the _____ by denying the infectious agent a susceptible host.
26. John is infected with influenza on April 8, first shows symptoms on April 10, and becomes contagious on April 12. In John's case, the _____ was 2 days and the _____ was 4 days.
27. _____ is a tool that allows researchers to assess the quality of evidence and the validity of public health recommendations that rely on the evidence.
28. _____ is a blanket term for a process which involves someone's immune system becoming protected against infection from an agent. One type of immunization involves injecting live, weakened microbes and is called _____. Another type of immunization is _____.
29. Bob reports experiencing symptoms such as fever, nausea, diarrhea, and abdominal pain. Over the next few days, several additional reports of similar symptoms are made. Thanks to _____ surveillance, the common symptoms lead to the discovery of a new outbreak. Because he was the first case within the outbreak that researchers discovered, Bob is the _____ of the outbreak.
30. Vaccines with a high _____ (the degree to which disease is prevented under ideal conditions) do not always have a high _____ because of the variability of conditions in the real world.
31. Jim O'Neill, U.S. deputy secretary of health and human services, succeeded Susan Monarez as acting director of the _____ in August 2025.
32. Mary Mallon was a _____ of typhoid fever because she was able to transmit the disease while infected but asymptomatic; mosquitoes are an example of a _____ because they can transmit malaria to humans without being infected with malaria themselves.
33. When a disease is _____, it is constantly present at some baseline level in a population. When this baseline gets very high, and the disease constantly has a large prevalence in the population, it is _____. When every individual in the population is infected, the disease is _____.
34. The first _____ was performed by Edward Jenner in 1796 to protect against smallpox.

PART 1 SCORE: _____ / 36

Part 2 | Matching

Match the disease to the type of agent that causes it. **[1 each]**

ANSWER WITH ONE OF THE FOLLOWING LETTERS: [**B**] Bacterium; [**V**] Virus; [**F**] Fungus; [**R**] Prion; [**S**] Parasite

35. Malaria
36. Trichinosis
37. Bubonic plague
38. Bovine spongiform encephalopathy (BSE)
39. COVID-19
40. Cryptosporidiosis
41. Herpes simplex
42. Creutzfeldt–Jakob disease
43. Salmonellosis
44. Diphtheria
45. Legionellosis
46. Histoplasmosis
47. Pneumocystis pneumonia
48. Pertussis
49. Anthrax

Which of the four types of surveillance is being used in each of the following? **[1 each]**

ANSWER WITH ONE OF THE FOLLOWING: Primary, secondary, tertiary, quaternary

50. An ischemic stroke victim attends rehab to improve his motor functions.
51. A woman receives mammograms at regular intervals to increase her odds of early detection of breast cancer.
52. A man wakes up with a sore throat and decides to take a lab test for strep throat.
53. A group distributes pamphlets and speaks at local schools about the risks of cigarette smoking.
54. A woman opts for a non-invasive pharmacological course of treatment rather than an invasive surgery with a long period of recovery.
55. A man with HIV infection takes one pill daily as part of an antiretroviral therapy (ART) regimen.
56. A woman is diagnosed with sinusitis and takes the antibiotic amoxicillin to treat the infection.

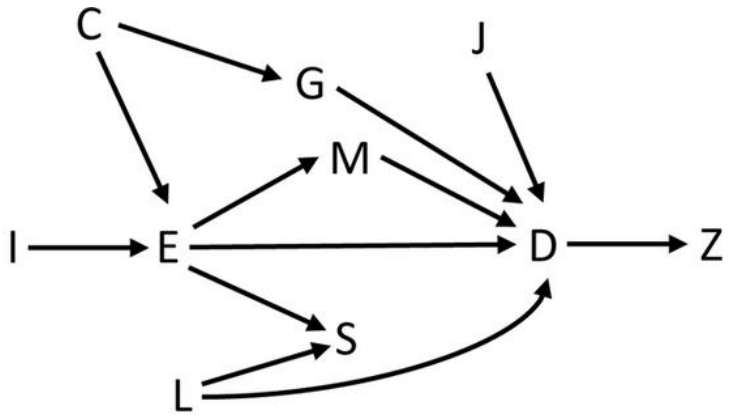
PART 2 SCORE: _____ / 22

Part 3 | General short answer

Fully answer each of the following using no more than three sentences.

57. What is the significance of a double-blind study design? [**2**]
58. Which study design is best suited for investigating a rare disease? Why? [**3**]
59. Why is herd immunity important? Why is a failure to attain herd immunity a risk? [**3**]
60. List the five phases of the public health surveillance process. [**2**]
61. Which of the main types of epi curve is most likely to be seen with a disease that is spread from person-to-person? Why? [**2**]
62. Under what conditions can a case be classified as "confirmed" without laboratory testing? [**2**]

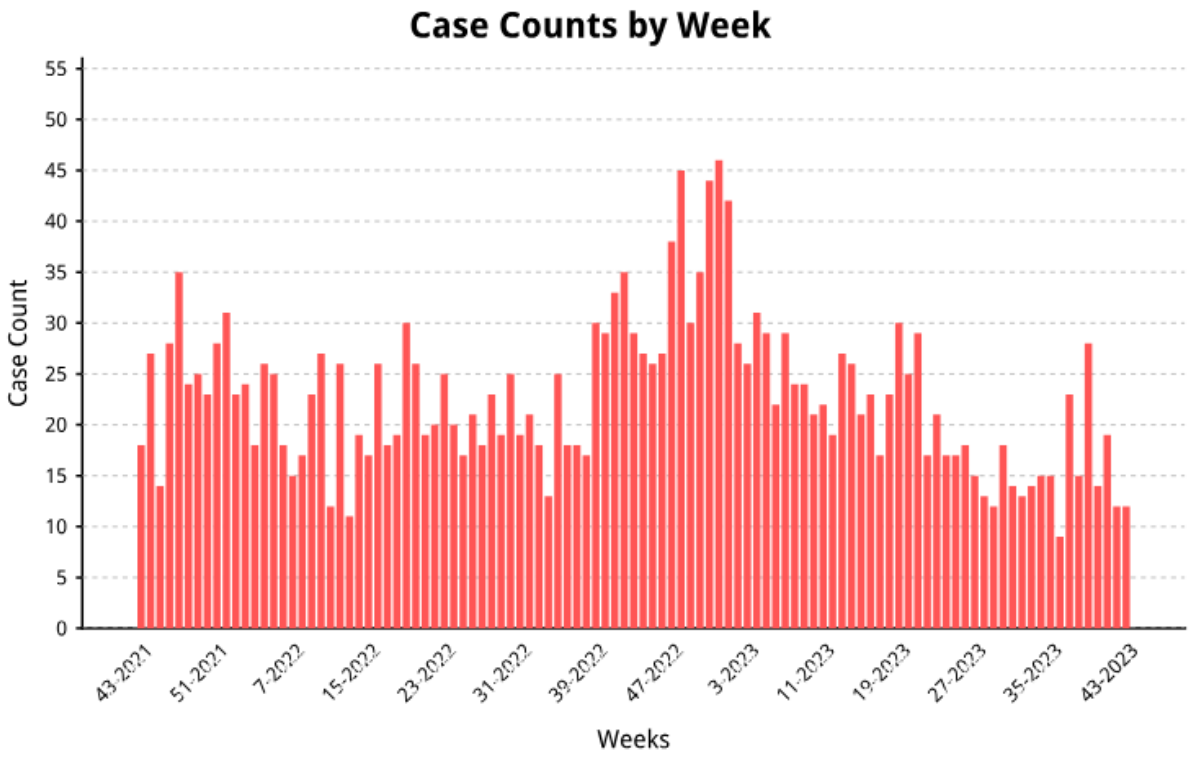
Answer the following questions about the directed acyclic graph (DAG) displayed below.[1]



63. Identify a causal path linking C and S, if one exists. [**1**]
64. Identify a causal path linking I and J, if one exists. [**1**]
65. Identify a causal path linking C and Z, if one exists. [**1**]
66. Identify a causal path linking J and E, if one exists. [**1**]
67. Identify one variable in the DAG that is acting as a collider in at least one path. [**1**]
68. Identify one variable in the DAG that is acting as an intermediary in at least one path. [**1**]
69. Which of the following <u>are</u> ways to block non-causal paths in a DAG? List all that apply. [**2**]
    a. Control for all mediators or intermediates on the path
    b. Control for all colliders on the path
    c. Do not control for any mediators or intermediates on the path
    d. Do not control for any colliders on the path

[1] Credit for the DAG visual: Figure 1 from Digitale JC, Martin JN, Glymour MM. Tutorial on directed acyclic graphs. *J Clin Epidemiol.* 2023:142:264–267. doi:10.1016/j.jclinepi.2021.08.001.

Answer the following questions about the epi curve displayed below showing cases of meningitis in Michigan.[2]



70. What is the full name for an epi curve? [**1**]
71. Which type of epi curve is shown above? [**1**]
72. What duration of time is displayed on this epi curve? [**1**]
73. According to the epi curve, onset of the first case took place during which week? [**1**]
74. The peak of cases took place during which week? [**1**]
75. What are the minimum and maximum number of cases reported in a given week? [**2**]
76. Suggest a better, more complete title for this epi curve. [**2**]

PART 3 SCORE: _____ / 31

[2] Credit for the epi curve visual: Weekly Disease Report for the Week Ending October 28, 2023 (MMWR Week 43). Michigan Disease Surveillance System, p. 32. https://www.michigan.gov/mdhhs/-/media/Project/Websites/mdhhs/CDINFO/WSR/WSR_43_2023.pdf

Part 4 | Calculation and application

Answer the following about this data gathered as part of a cohort study.

	Exposure	No exposure	TOTAL
Disease	74	29	**103**
No disease	31	103	**134**
TOTAL	**105**	**132**	**237**

77. What is the appropriate measure of association for this study? [**1**]
78. Perform the calculation you identified in Q77. [**2**]
79. Put your answer to Q78 in context. What does that number mean? [**2**]
80. Your colleague looks at your calculations and says that the exposure is the likely cause of the disease. Do you agree with him? Why? [**3**]

Answer the following about this data gathered as part of a cross-sectional study.

	Exposure	No exposure	TOTAL
Disease	173	72	**245**
No disease	88	99	**187**
TOTAL	**261**	**171**	**432**

81. What is the prevalence of disease in the study population? [**2**]
82. What is the attack rate for the exposure? [**1**]
83. Can you conclude that the exposure was the cause of the disease? Justify your answer. [**3**]

Answer the following about this data gathered as part of a case-control study.

	Exposure A	Exposure B	No exposure	TOTAL
Disease	116	124	17	**257**
No disease	162	90	308	**560**
TOTAL	**278**	**214**	**325**	**817**

84. What is the appropriate measure of association for this study? [**1**]
85. Perform the calculation you identified in Q84 for both exposures. [**4**]
86. Which exposure is more likely to have caused the disease? Justify your answer. [**3**]

PART 4 SCORE: _____ / 22